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parameter : N = 20, a = 2, b = 0, n = 3, theta θ = 60°, alpha = 0.001

tanggal : 2026 - 4 - 20

(1.) dik :

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} \xrightarrow{\text{memasukkan parameter.}} \begin{bmatrix} 2+1 & 0 & 2 \\ 1 & 2+2 & 0 \\ 0 & 1 & 2+3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 \\ 1 & 4 & 0 \\ 0 & 1 & 5 \end{bmatrix}$$

(a.) matriks eksplisit dit : det (A) ?

$$A = \begin{bmatrix} 3 & 0 & 2 \\ 1 & 4 & 0 \\ 0 & 1 & 5 \end{bmatrix}$$

det (A) ekspansi kofaktor. (baris 1)

$|A| = a_{11} + C_{11} + a_{12} C_{12} + a_{13} + C_{13}$

$C_{ij} = (-1)^{i+j} \cdot M_{ij}$

→ kofaktor C_{11}

$$C_{11} = + \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix} = (4.5) - (0.1) = 20$$

$$C_{12} = - \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} = -((1.5) - (0.0)) = -5$$

$$C_{13} = + \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = (1.1) - (4.0) = 1$$

$$\det(A) = (3.20) + (0.-5) + (2.1)$$
$$= 62$$

(b.) matriks minor

$$M_{11} = \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix} = (4.5) - (0.1) = 20$$

$$M_{12} = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} = (1.5) - (0.0) = 5$$

$$M_{13} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = (1.1) - (4.0) = 1$$

$$M_{21} = \begin{bmatrix} 0 & 2 \\ 1 & 5 \end{bmatrix} = (0.5) - (2.1) = -2$$

$$M_{22} = \begin{bmatrix} 3 & 2 \\ 0 & 5 \end{bmatrix} = (3.5) - (2.0) = 15$$

$$M_{23} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = 3 - 0 = 3$$

$$M_{31} = \begin{bmatrix} 0 & 2 \\ 4 & 0 \end{bmatrix} = 0 - 8 = -8$$

$$M_{32} = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix} = 0 - 2 = -2$$

$$M_{33} = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} = 12 - 0 = 12$$

$$M = \begin{bmatrix} 20 & 5 & 1 \\ -2 & 15 & 3 \\ -8 & -2 & 12 \end{bmatrix}$$

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(c.) dit :

→ matrus kofaktor (C)

→ adjoin (adj (A))

→ Invers (A^{-1})

MATRISK KOFAKTOR ($C_{ij} = (-1)^{i+j} \cdot M_{ij}$)

$C_{11} = (-1)^{1+1} \cdot M_{11} = 20$ $\quad C_{21} = (-1)^{2+1} \cdot M_{21} = 2$ $\quad C_{31} = (-1)^{3+1} \cdot M_{31} = -8$

$C_{12} = (-1)^{1+2} \cdot M_{12} = -5$ $\quad C_{22} = (-1)^{2+2} \cdot M_{22} = 15$ $\quad C_{32} = (-1)^{3+2} \cdot M_{32} = 2$

$C_{13} = (-1)^{1+3} M_{13} = 1$ $\quad C_{23} = (-1)^{2+3} \cdot M_{23} = -3$ $\quad C_{33} = (-1)^{3+3} \cdot M_{33} = 12$

$$C = \begin{bmatrix} 20 & -5 & 1 \\ 2 & 15 & -3 \\ -8 & 2 & 12 \end{bmatrix}$$

MATRIKS ADJOIN ($adj(A) = C^T$)

$$adj(A) = C^T = \begin{bmatrix} 20 & 2 & -8 \\ -5 & 15 & 2 \\ 1 & -3 & 12 \end{bmatrix}$$

MATRIKS INVERS ($A^{-1} = \frac{1}{det(A)} \cdot adj(A)$)

dik :

$det(A) = 62$

$$adj(A) = \begin{bmatrix} 20 & 2 & -8 \\ -5 & 15 & 2 \\ 1 & -3 & 12 \end{bmatrix}$$

Jawab

$$A^{-1} = \frac{1}{62} \cdot \begin{bmatrix} 20 & 2 & -8 \\ -5 & 15 & 2 \\ 1 & -3 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 20/62 & 2/62 & -8/62 \\ -5/62 & 15/62 & 2/62 \\ 1/62 & -3/62 & 12/62 \end{bmatrix} \text{ atau } \begin{bmatrix} 10/31 & 1/31 & -4/31 \\ -5/62 & 15/62 & 1/31 \\ 1/62 & -3/62 & 6/31 \end{bmatrix}$$

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(d) [MATLAB]

(e) dik :

blok pixel (mewakili sudut logo)

$$P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 70 & 200 \\ 80 & 4 & 30 \\ 50 & 90 & 170 \end{bmatrix}$$

dit :

→ enkripsi $E = A \cdot P$

→ dekripsi $P' = A^{-1} \cdot E$

→ buktikan $P' = P$

→ apa yang terjadi jika $\det(A) = 0$?

Jawab :

ENKRIPSI $E = A \cdot P$

$$E = \begin{bmatrix} 3 & 0 & 2 \\ 1 & 4 & 0 \\ 0 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 120 & 70 & 200 \\ 80 & 4 & 30 \\ 50 & 90 & 170 \end{bmatrix}$$

$$E = \begin{bmatrix} 3(120) + 0 + 2(50) & 3(70) + 0 + 2(90) & 3(200) + 0 + 2(170) \\ 1(120) + 4(80 + 0 & 1(70) + 4(4) + 0 & 1(200) + 4(30) + 0 \\ 0(120) + 1(80) + 5(50 & 0 + 1(4) + 5(90) & 0 + 1(30) + 5(170) \end{bmatrix}$$

$$E = \begin{bmatrix} 360 + 0 + 100 & 210 + 180 & 600 + 340 \\ 120 + 320 + 0 & 70 + 16 & 200 + 120 \\ 0 + 80 + 250 & 4 + 450 & 30 + 850 \end{bmatrix}$$

$$E = \begin{bmatrix} 460 & 390 & 940 \\ 450 & 86 & 320 \\ 330 & 454 & 880 \end{bmatrix}$$

matrius E diatas adalah hasil enkripsi dari blok citra P. terlihat bahwa nilai2x pixel telah berubah drastis menjadi lebih besar, inilah yang menyebabkan gambar aslinya tidak dapat dikenali secara visual tanpa proses dekripsi menggunakan A^{-1}

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DEKRIPSI $P' = A^{-1} \cdot E$ + pembuktian $P' = P$

$$P' = \frac{1}{6}\begin{bmatrix} 20 & 2 & -8 \\ -5 & 15 & 2 \\ 1 & -3 & 12 \end{bmatrix} \cdot \begin{bmatrix} 460 & 390 & 940 \\ 440 & 86 & 320 \\ 330 & 454 & 880 \end{bmatrix}$$

$$P' = \frac{1}{6}\begin{bmatrix} (20.460)+(2.440)+(-8.330) & 20(390)+(2.86)+(-8.454) & 20(940)+2(320)+(-8.880) \\ (-5.460)+(15.440)+(2.330) & (-5(390)+(15.86)+(2.454) & -5(940)+(15.320)+(2.880) \\ (1.460)+(-3.440)+(12.330) & (1.390)+(-3.86)+(12.454) & (1.940)+(-3.320)+(12.880) \end{bmatrix}$$

$$P' = \frac{1}{6}\begin{bmatrix} 9200+880-2640 & 7800+172-3632 & 18800+640-7040 \\ -2300+6600+660 & -1950+1290+908 & -9700+4800+1760 \\ 460-1320+3960 & 390-258+5448 & 940-960+10560 \end{bmatrix}$$

$$P' = \frac{1}{6}\begin{bmatrix} 7440 & 4340 & 12400 \\ 4960 & 248 & 1860 \\ 3100 & 5580 & 10540 \end{bmatrix} = \begin{bmatrix} 120 & 70 & 200 \\ 80 & 4 & 30 \\ 50 & 90 & 170 \end{bmatrix} = P = \begin{bmatrix} 120 & 70 & 200 \\ 80 & 4 & 30 \\ 50 & 90 & 170 \end{bmatrix}$$

[TERBUKTI $P' = P$]

pembuktian singkat ↓

dik: $E = A \cdot P$; $P' = A' \cdot E$

$\frac{E}{A} = P$

$A' \cdot E = P$

$P = A' \cdot E = P'$ //

(f.) [MATLAB]

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[lanjutan] e

jika det (A) = 0, maka matrius merupakan matrias singular, sehingga matrius A tidak memiliki invers. sehingga matrius tetap bisa di enkripsi, namun tidak bisa dikembalikan lagi / dekripsi. karena kunci dekripsi adalah mengalikannya dengan invers.

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